DataAnalysis

March 7, 2026

[2]:
```
import os
import glob
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("seaborn-v0_8")
```

[3]:
```
wifi_path = "Datasets/WiFi dataset"
mag_path = "Datasets/Magnetic field dataset"

print("WiFi folder exists:", os.path.exists(wifi_path))
print("Magnetic folder exists:", os.path.exists(mag_path))
```

```
WiFi folder exists: True
Magnetic folder exists: True
```

[4]:
```
wifi_files = glob.glob("Datasets/WiFi dataset/**/*.csv", recursive=True)
mag_files = glob.glob("Datasets/Magnetic field dataset/**/*.csv",␣
 ↪recursive=True)

print("Total WiFi files:", len(wifi_files))
print("Total Magnetic files:", len(mag_files))
```

```
Total WiFi files: 2831
Total Magnetic files: 4261
```

[5]:
```
sample_wifi_file = wifi_files[0]
print(sample_wifi_file)
```

```
Datasets/WiFi dataset\BE Engineering\Navigation\Scenario-1\A8\User 2\WiFi_BE
Building_Scenario 1_User 2 (M-174cm)_Navigation_2021.05.19 171520.csv
```

[6]:
```
with open(sample_wifi_file, "rb") as f:
    print(f.read(200))
```

```
b'\xd0\xcf\x11\xe0\xa1\xb1\x1a\xe1\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x
00\x00\x00\x00\x00>\x00\x03\x00\xfe\xff\t\x00\x06\x00\x00\x00\x00\x00\x00\x00\x0
```

```
0\x00\x00\x00\x01\x00\x00\x007\x00\x00\x00\x00\x00\x00\x00\x00\x10\x00\x00\xfe\x
ff\xff\xff\x00\x00\x00\x00\xfe\xff\xff\xff\x00\x00\x00\x006\x00\x00\x00\xff\xff\
xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\
xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\
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xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\
xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff\xff'
```

[7]:
```
df_wifi = pd.read_excel(sample_wifi_file)

df_wifi.head()
```

[7]:

	Time	X-pos	Y-pos	SSID	BSSID	RSS
0	2021:05:19 17:15:20	1.0	24.0	WAP5089	00:1a:be:a0:de:89	-53
1	NaN	NaN	NaN	WAP5144	00:1a:be:a0:df:45	-53
2	NaN	NaN	NaN	WAP5146	00:1a:be:a0:df:47	-56
3	NaN	NaN	NaN	WAP5091	00:1a:be:a0:de:91	-57
4	NaN	NaN	NaN	WAP5090	00:1a:be:a0:de:90	-61

[8]:
```
df_wifi.shape
```

[8]: (37, 6)

[9]:
```
df_wifi.columns
```

[9]: Index(['Time', 'X-pos', 'Y-pos', 'SSID', 'BSSID', 'RSS'], dtype='object')

[10]:
```
df_wifi.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 37 entries, 0 to 36
Data columns (total 6 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Time    1 non-null      object
 1   X-pos   1 non-null      float64
 2   Y-pos   1 non-null      float64
 3   SSID    37 non-null     object
 4   BSSID   37 non-null     object
 5   RSS     37 non-null     int64
dtypes: float64(2), int64(1), object(3)
memory usage: 1.9+ KB
```

[11]:
```
df_wifi.isnull().sum()
```

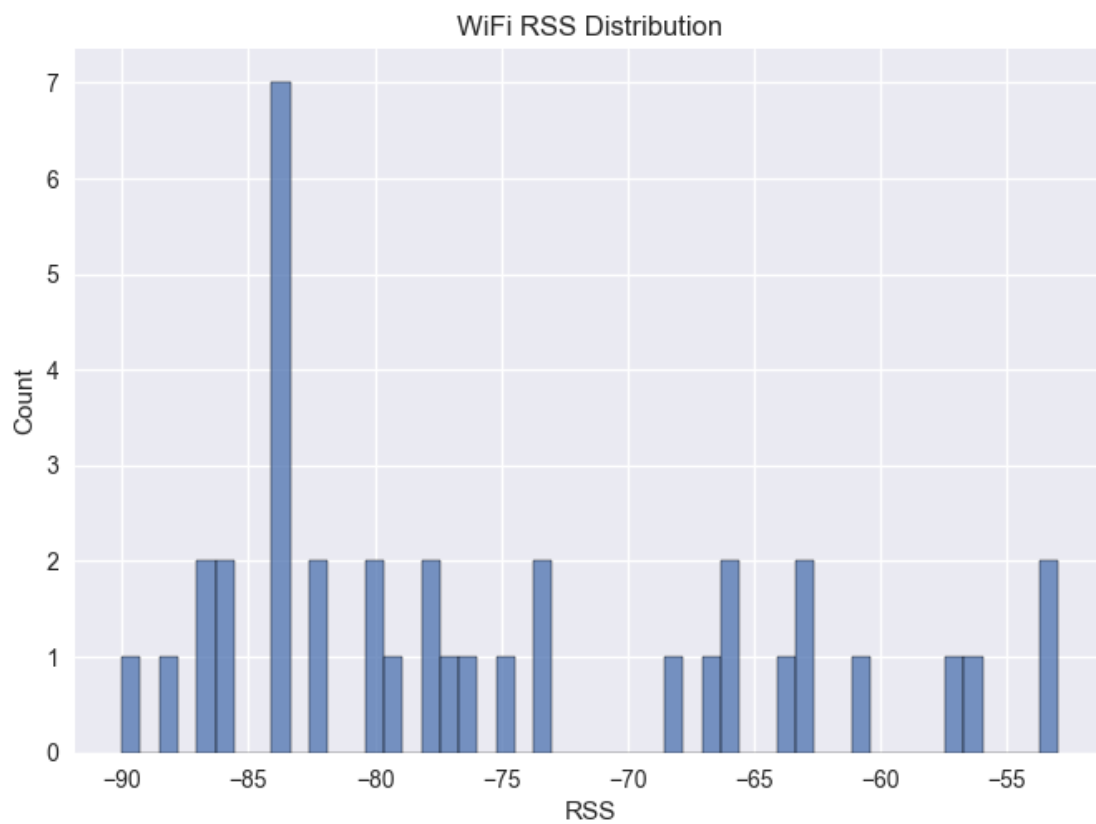
[11]:
```
Time     36
X-pos    36
Y-pos    36
```

```
SSID      0
BSSID     0
RSS       0
dtype: int64
```

[12]:
```
df_wifi["RSS"].describe()
```

[12]:
```
count    37.000000
mean    -75.189189
std      10.823939
min     -90.000000
25%     -84.000000
50%     -78.000000
75%     -66.000000
max     -53.000000
Name: RSS, dtype: float64
```

[13]:
```
sns.histplot(df_wifi["RSS"], bins=50)
plt.title("WiFi RSS Distribution")
plt.show()
```



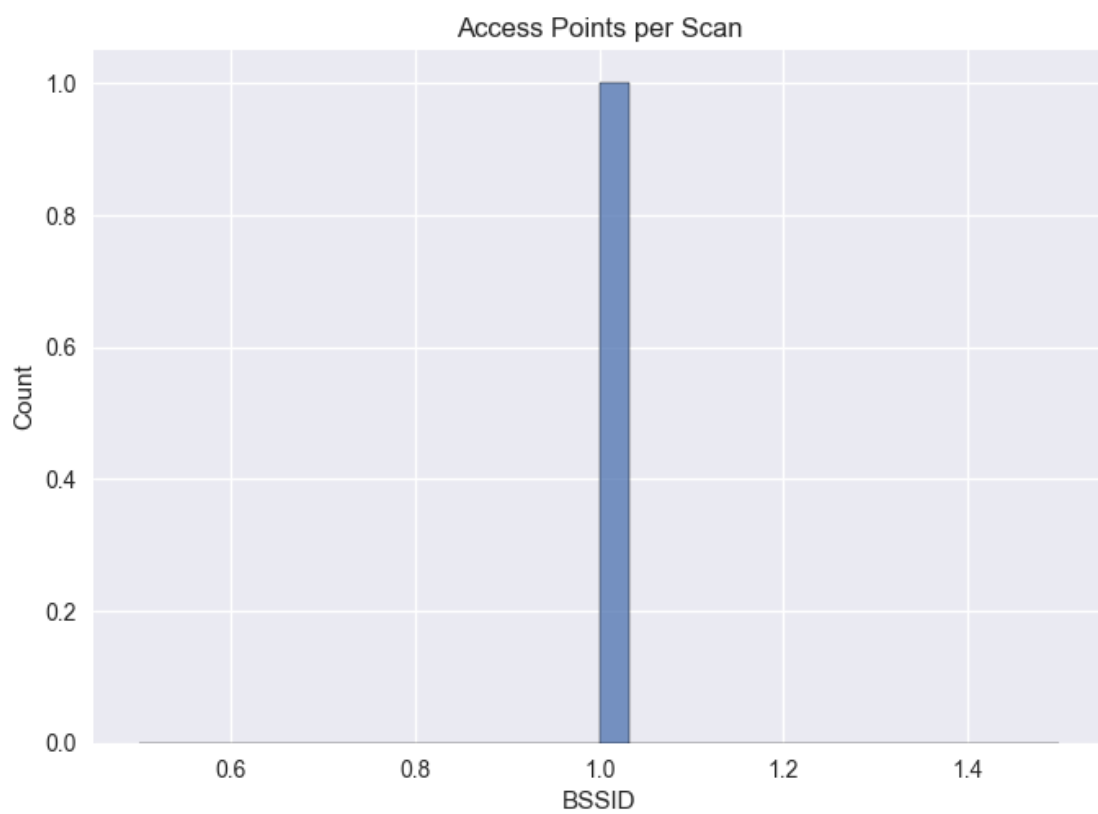
[14]:
```
df_wifi["BSSID"].nunique()
```

[14]: 37

[15]:
```
ap_counts = df_wifi.groupby("Time")["BSSID"].count()

ap_counts.describe()
```

[15]:
```
count    1.0
mean     1.0
std      NaN
min      1.0
25%      1.0
50%      1.0
75%      1.0
max      1.0
Name: BSSID, dtype: float64
```

[16]:
```
sns.histplot(ap_counts, bins=30)
plt.title("Access Points per Scan")
plt.show()
```



[17]:
```
sample_mag_file = mag_files[0]

print(sample_mag_file)
```

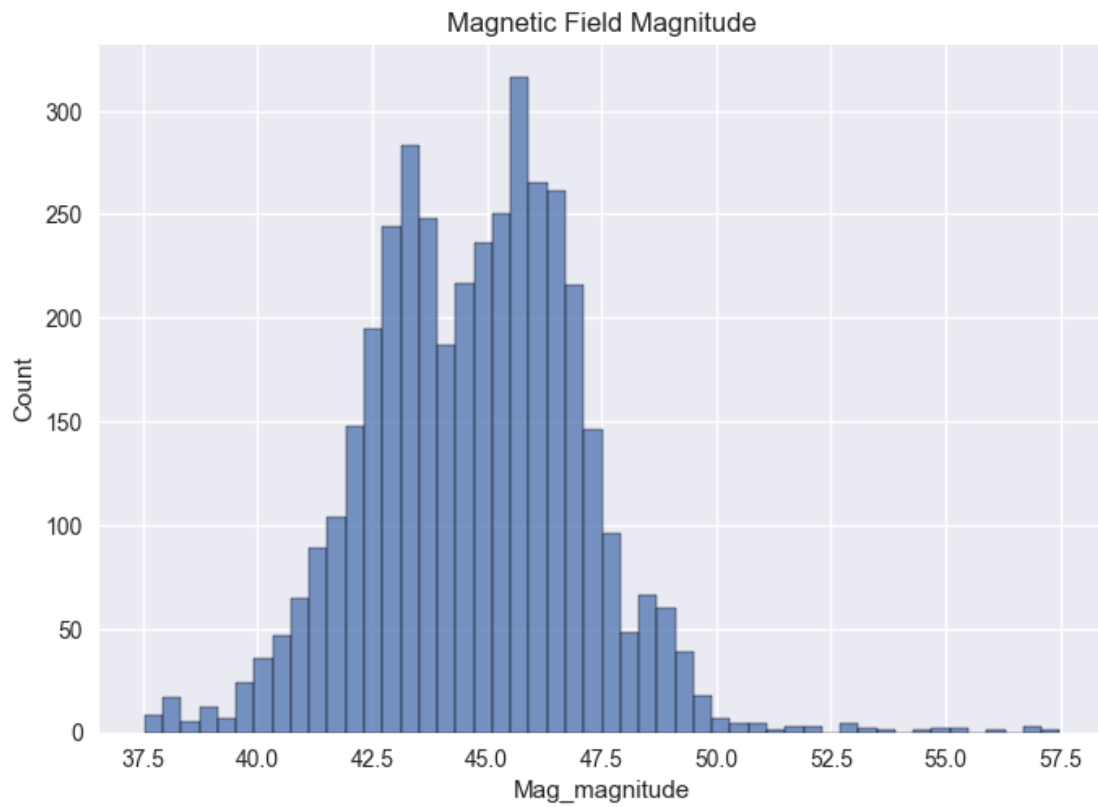
```
Datasets/Magnetic field dataset\Continuous Data\BE Building\Navigation\Scenario
1\A8\User 2\IMU_BE Building_Scenario 1_User 2 (M-174cm)_Navigation_2021.05.19
173743.csv
```

[18]:
```
df_mag = pd.read_csv(sample_mag_file)

df_mag.head()
```

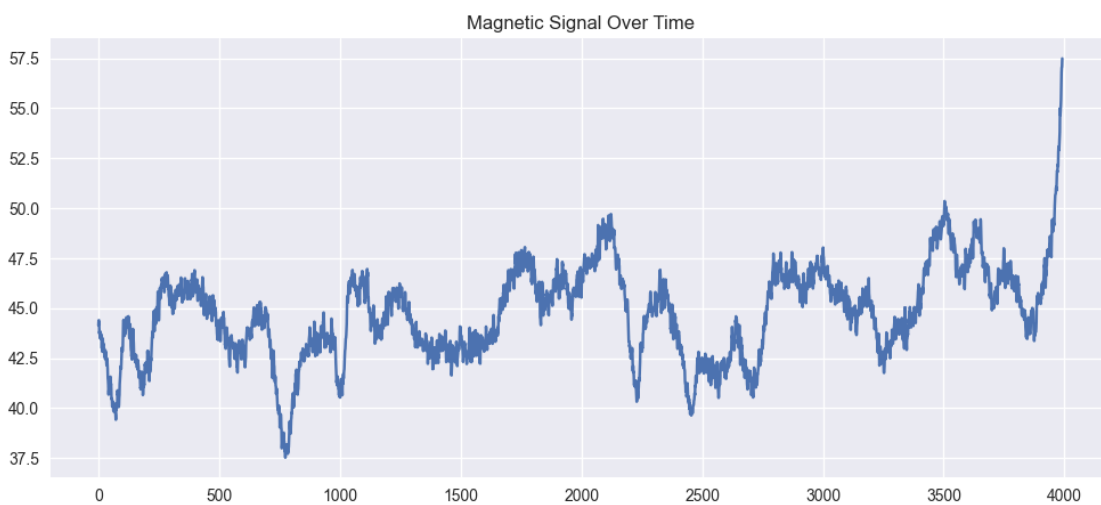
[18]:
```
                  Time      Mag_x      Mag_y      Mag_z     Acc_x     Acc_y  \
0  2021.05.19 17:37:43  32.820000 -13.320000 -26.340000 -0.481235  0.931344
1  2021.05.19 17:37:43  33.000000 -13.559999 -26.400000  0.090980  1.331176
2  2021.05.19 17:37:43  33.120000 -13.440000 -26.340000 -0.538695  1.290475
3  2021.05.19 17:37:43  33.059998 -13.200000 -26.099998 -0.507571  1.359907
4  2021.05.19 17:37:43  32.700000 -13.679999 -25.619999 -0.632069  1.661576

      Acc_z    Gyro_x    Gyro_y    Gyro_z     Orn_x     Orn_y     Orn_z  \
0  9.543288  0.003665  0.090408  0.103847 -0.097161  0.050384 -1.894290
1  9.612720 -0.076969 -0.136223  0.218079 -0.137599 -0.009464 -1.858915
2  9.311050 -0.001222 -0.064141  0.133779 -0.137492  0.057791 -1.861784
3  9.502586  0.015272 -0.153327  0.080023 -0.141944  0.053363 -1.851907
4  9.392453 -0.007330 -0.143553  0.054978 -0.174707  0.067194 -1.843602

     Pressure
0  1004.03784
1  1004.03784
2  1004.03784
3  1004.09230
4  1004.09230
```

[19]:
```
df_mag["Mag_magnitude"] = np.sqrt(
    df_mag["Mag_x"]**2 +
    df_mag["Mag_y"]**2 +
    df_mag["Mag_z"]**2
)
```

[20]:
```
sns.histplot(df_mag["Mag_magnitude"], bins=50)
plt.title("Magnetic Field Magnitude")
plt.show()
```



[21]:
```
df_mag["Mag_magnitude"].plot(figsize=(12,5))
plt.title("Magnetic Signal Over Time")
plt.show()
```



[22]:
```
wifi_root = "Datasets/WiFi dataset"

wifi_files = []

for root, dirs, files in os.walk(wifi_root):
    for file in files:
        if file.endswith(".csv"):
            wifi_files.append(os.path.join(root, file))

print("Total WiFi files:", len(wifi_files))
```

```
Total WiFi files: 2831
```

[23]:
```
wifi_files[:5]
```

[23]:
```
['Datasets/WiFi dataset\\BE Engineering\\Navigation\\Scenario-1\\A8\\User
2\\WiFi_BE Building_Scenario 1_User 2 (M-174cm)_Navigation_2021.05.19
171520.csv',
 'Datasets/WiFi dataset\\BE Engineering\\Navigation\\Scenario-1\\A8\\User
2\\WiFi_BE Building_Scenario 1_User 2 (M-174cm)_Navigation_2021.05.19
171534.csv',
 'Datasets/WiFi dataset\\BE Engineering\\Navigation\\Scenario-1\\A8\\User
2\\WiFi_BE Building_Scenario 1_User 2 (M-174cm)_Navigation_2021.05.19
171547.csv',
 'Datasets/WiFi dataset\\BE Engineering\\Navigation\\Scenario-1\\A8\\User
2\\WiFi_BE Building_Scenario 1_User 2 (M-174cm)_Navigation_2021.05.19
171602.csv',
 'Datasets/WiFi dataset\\BE Engineering\\Navigation\\Scenario-1\\A8\\User
2\\WiFi_BE Building_Scenario 1_User 2 (M-174cm)_Navigation_2021.05.19
171732.csv']
```

[24]:
```
ap_set = set()

for file in wifi_files[:100]:   # start small
    df = pd.read_excel(file)
    ap_set.update(df["BSSID"].unique())

print("Unique APs across dataset:", len(ap_set))
```

```
Unique APs across dataset: 164
```

[25]:
```
rss_values = []

for file in wifi_files[:200]:
    df = pd.read_excel(file)
    rss_values.extend(df["RSS"].values)

print("RSS min:", min(rss_values))
```

```
print("RSS max:", max(rss_values))
```

```
RSS min: -95
RSS max: 0
```

[26]:
```
zero_count = 0
total_count = 0

for file in wifi_files[:200]:
    df = pd.read_excel(file)
    zero_count += (df["RSS"] == 0).sum()
    total_count += len(df)

print("Zero RSS count:", zero_count)
print("Total RSS values:", total_count)
print("Zero percentage:", zero_count / total_count)
```

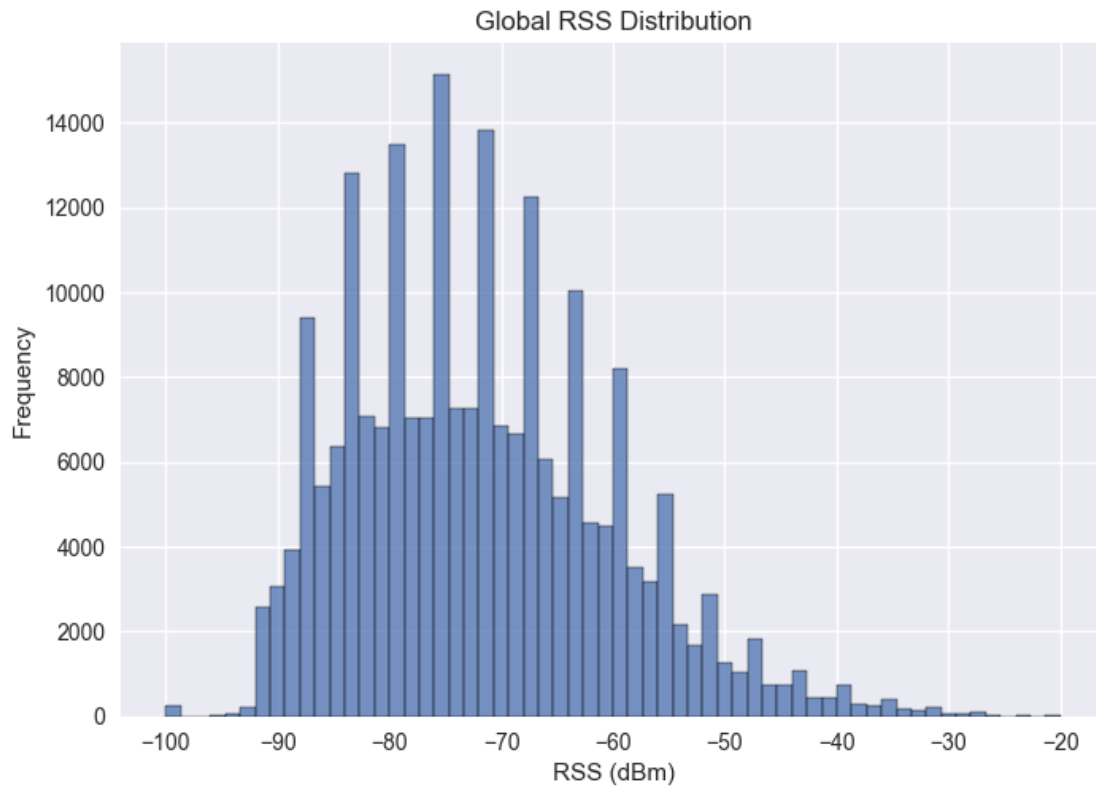
```
Zero RSS count: 6
Total RSS values: 7186
Zero percentage: 0.0008349568605622043
```

[27]:
```
df["RSS"] = df["RSS"].replace(0, -100)
```

[28]:
```
rss_all = []

for file in wifi_files:
    df = pd.read_excel(file)
    rss_all.extend(df["RSS"].replace(0, -100).values)

rss_all = pd.Series(rss_all)

print(rss_all.describe())
```

```
count    221999.000000
mean        -71.881563
std          11.565214
min        -100.000000
25%         -81.000000
50%         -73.000000
75%         -64.000000
max         -20.000000
dtype: float64
```

[29]:
```
sns.histplot(rss_all, bins=60)
plt.title("Global RSS Distribution")
plt.xlabel("RSS (dBm)")
plt.ylabel("Frequency")
plt.show()
```



[30]:
```
from collections import Counter

ap_counter = Counter()

for file in wifi_files:
    df = pd.read_excel(file)
    ap_counter.update(df["BSSID"])

print("Total APs:", len(ap_counter))
```

```
Total APs: 1425
```

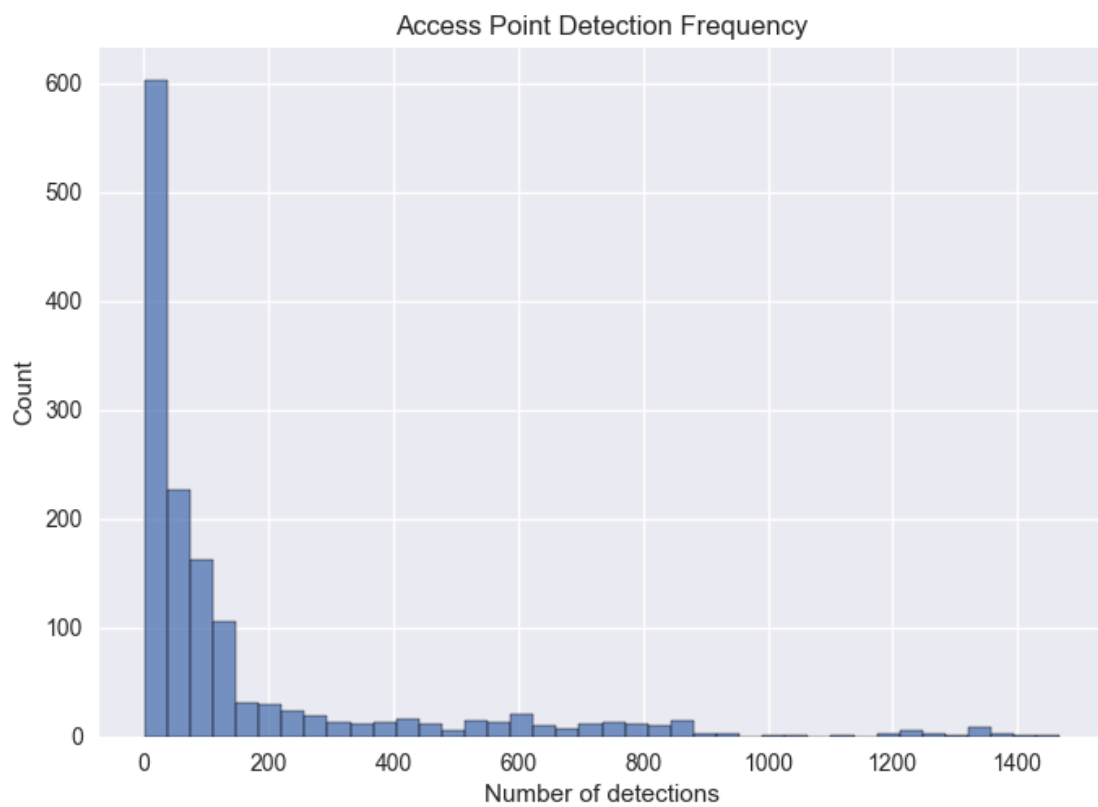
[31]:
```
ap_freq = pd.DataFrame(
    ap_counter.items(),
    columns=["BSSID","count"]
).sort_values("count", ascending=False)

ap_freq.head()
```

[31]:

	BSSID	count
964	00:1a:ac:a0:de:s6	1468
974	00:1a:ac:a0:de:s7	1418

```
961    00:1a:ac:a0:de:s4    1367
1006   00:1a:ac:a0:de:s1    1366
972    00:1a:ac:a0:de:s5    1365
```

[32]:
```
sns.histplot(ap_freq["count"], bins=40)
plt.title("Access Point Detection Frequency")
plt.xlabel("Number of detections")
plt.show()
```

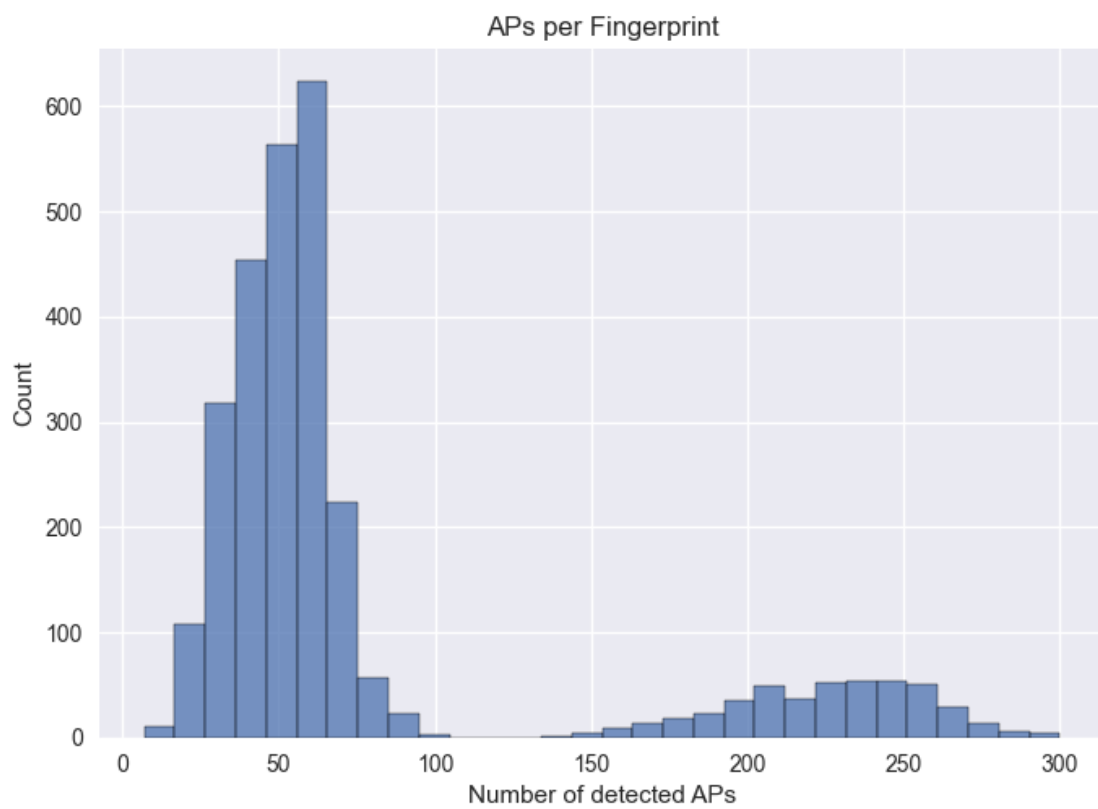


[33]:
```
ap_counts = []

for file in wifi_files:
    df = pd.read_excel(file)
    ap_counts.append(len(df))

ap_counts = pd.Series(ap_counts)

print(ap_counts.describe())
```

```
count    2831.000000
mean       78.417167
std        66.542668
```

```
min          7.000000
25%         43.000000
50%         55.000000
75%         67.000000
max        300.000000
dtype: float64
```

[34]:
```
sns.histplot(ap_counts, bins=30)
plt.title("APs per Fingerprint")
plt.xlabel("Number of detected APs")
plt.show()
```



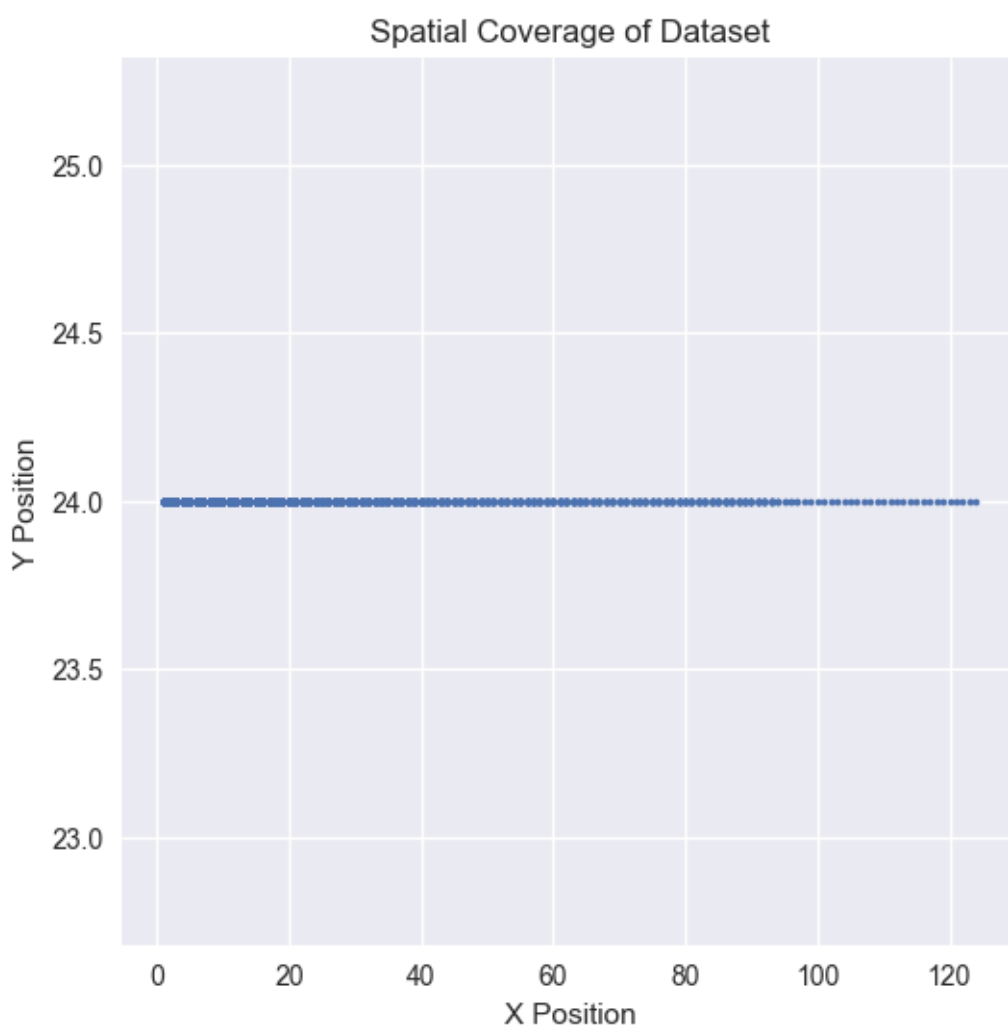
[38]:
```
x_positions = []
y_positions = []

for file in wifi_files:

    df = pd.read_excel(file)

    if "X-pos" in df.columns and "Y-pos" in df.columns:
```

```
        x_vals = df["X-pos"].dropna()
        y_vals = df["Y-pos"].dropna()

        if len(x_vals) > 0 and len(y_vals) > 0:
            x_positions.append(x_vals.iloc[0])
            y_positions.append(y_vals.iloc[0])
```

[39]:
```
print("Total labeled fingerprints:", len(x_positions))
```

```
Total labeled fingerprints: 2381
```

[40]:
```
plt.figure(figsize=(6,6))
plt.scatter(x_positions, y_positions, s=5)
plt.title("Spatial Coverage of Dataset")
plt.xlabel("X Position")
plt.ylabel("Y Position")
plt.show()
```



[41]:
```
column_sets = set()

for file in wifi_files[:100]:
    df = pd.read_excel(file)
    column_sets.add(tuple(df.columns))

column_sets
```

[41]:
```
{('Time', 'X-pos', 'Y-pos', 'SSID', 'BSSID', 'RSS')}
```

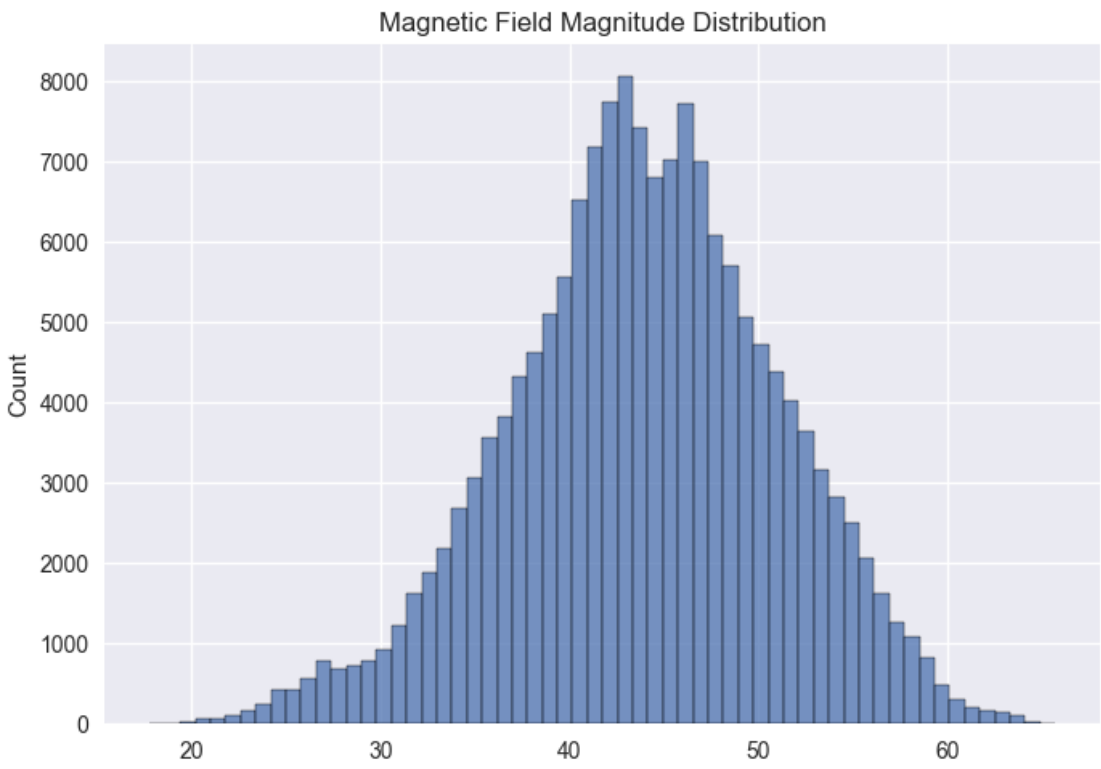
[42]:
```
print(df.columns)
```

```
Index(['Time', 'X-pos', 'Y-pos', 'SSID', 'BSSID', 'RSS'], dtype='object')
```

[43]:
```
mag_values = []

for file in mag_files[:200]:  # start small
    df = pd.read_csv(file)

    mag = (df["Mag_x"]**2 + df["Mag_y"]**2 + df["Mag_z"]**2)**0.5
    mag_values.extend(mag)

mag_values = pd.Series(mag_values)

print(mag_values.describe())
```

```
count    161397.000000
mean         43.905758
std           7.114145
min          17.803119
25%          39.410227
50%          43.948229
75%          48.701450
max          65.722161
dtype: float64
```

[44]:
```
sns.histplot(mag_values, bins=60)
plt.title("Magnetic Field Magnitude Distribution")
plt.show()
```


Magnetic Field Magnitude Distribution
Count
8000
7000
6000
5000
4000
3000
2000
1000
0
20
30
40
50
60